

Ectomycorrhizal fungal community assembly on seedlings of a Neotropical monodominant tree

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Abstract

Ectomycorrhizal tree species may benefit from positive plant–soil feedbacks, where soil environments near adult trees enhance conspecific seedling growth and survival. In tropical monodominant forests, seedling survival is particularly important, as seedling banks help maintain stand-level dominance over generations. Positive plant–soil feedbacks may be mediated by diverse ectomycorrhizal fungal communities, which improve nutrient acquisition of heavily shaded seedlings. Despite the potential importance of these fungi, little is known about ectomycorrhizal fungal community development on seedlings of tropical monodominant trees. In Guyana, we sequentially monitored percent colonization and species composition of ectomycorrhizal fungi on an even-age cohort of seedlings of the tropical monodominant tree *Dicymbe corymbosa* (Fabaceae subfamily Detarioideae). Ectomycorrhizal fungi found on *D. corymbosa* seedlings over a 12-month period of early development were compared to those of conspecific adults and four other ectomycorrhizal tree species in the region. Species turnover was high (80%) between 6- and 12-month-old seedlings, though the */russula-lactarius*, */clavulina*, and */tomentella-thelephora* lineages were species-rich on seedlings at all ages. The number of ectomycorrhizal morphotypes per seedling increased with age, but extent of fungal colonization did not. Seedling ectomycorrhizal fungi were shared with sympatric conspecific adults (55%) and, to a lesser extent, regional heterospecific adults (27%), but numerous species were previously unrecorded for Guyana. Over their development *D. corymbosa* seedlings did not rely strictly on adult trees for their mycobionts but appeared to foster unique assemblages of ectomycorrhizal fungi.

KEYWORDS

Clavulinaceae, Detarioideae, ectomycorrhizas, fungal community sequencing, Guiana Shield, Russulaceae, Thelephoraceae, tropical monodominance

1 | INTRODUCTION

Monodominance is common in temperate and boreal ecosystems but less common in the tropics (ter Steege et al., 2019). Most lowland tropical forests are very tree diverse (Richards, 1996), but stands occur in which a single tree species is dominant (e.g., Alexander & Högberg, 1986; Corrales et al., 2016; Henkel, 2003). These tropical monodominant (TMD) forests are known from the Neotropics (e.g., Corrales et al., 2016; Henkel, 2003), Guineo-Congolian Africa (e.g., Hall et al., 2020; Hart et al., 1989; Kearsley et al., 2017), and SE Asia (e.g., Grellier et al., 1987; Itoh et al., 1997; van der Velden et al., 2014). In monodominant stands, alpha diversity of trees is low by tropical standards, and the dominant species comprises 60% or more of the canopy trees (Connell & Lowman, 1989).

Nearly all TMD tree species associate with root symbiotic ectomycorrhizal (ECM) fungi (Corrales et al., 2018; ter Steege et al., 2019). Like TMD forests, the ECM mutualism is patchily distributed in the tropics, as most tropical woody angiosperms form arbuscular mycorrhizal (AM) symbioses (Tedersoo & Brundrett, 2017). A functional connection between the ECM symbiosis and TMD has long been hypothesized (e.g., Connell & Lowman, 1989; Janzen, 1974). Proposed mechanisms have centered on the superior ability of ECM fungi to recycle litter nutrients relative to AM fungi (see Courty et al., 2010; Phillips et al., 2013; Smith & Read, 2008, and references therein), an advantage that is evident on a genetic level (Miyachi et al., 2020). Competitive exclusion of AM trees may facilitate dominance by ECM trees in both low- and high-latitude ecosystems (Corrales et al., 2018; Finlay & Read, 1986; Henkel et al., 2005; Hobbie & Colpaert, 2003; Mayor et al., 2009).

In addition to forming ectomycorrhizas, many TMD trees exhibit supra-annual mast seeding that results in establishment of dense, heavily shaded seedling cohorts under parental adults (e.g., Henkel & Mayor, 2019; Torti et al., 2001). In tree-diverse tropical rainforests, seedlings of AM tree species have low survival rates when near their parent tree or in high conspecific densities (Comita et al., 2014; Connell, 1971; Janzen, 1970). This is a consequence of a type of negative plant-soil feedback (PSF) involving species-specific pathogens in the soil environment of adult trees that inhibit the growth and survival of conspecific seedlings (Bagchi et al., 2014; Mangan et al., 2010). In TMD stands of ECM trees, however, many seedlings establish under parent trees with subsequent low mortality rates (e.g., Henkel & Mayor, 2019). Seedling survival in such TMD stands may be a consequence of a positive PSF (Klironomos, 2002) in which the soil environment near adult trees, along with their ECM fungal symbionts, may benefit conspecific seedlings (Putten et al., 2016). Plant-soil feedbacks, whether positive or negative, are major influencers of tree community composition (van der Heijden et al., 2008; Mangan et al., 2010). The establishment of shade-tolerant seedlings in TMD forests could also indicate a dampening of negative PSFs (Connell & Lowman, 1989; Hart, 1990; Henkel & Mayor, 2019; Read et al., 1995).

One of the ways in which adult TMD trees may positively impact conspecific seedlings is through their influence on the ECM

mycobiota. In a Bornean forest co-dominated by multiple species of ECM Dipterocarpaceae, ECM seedlings experienced positive growth responses and increased mycorrhization when grown in the soil of closely related adults (Segnitz et al., 2020). Ectomycorrhizal fungi may also benefit seedlings by linking them with conspecific adult trees in a mycorrhizal network, through which seedlings may access adult resource pools and have improved chances of survival (Booth & Hoeksema, 2010; Corrales et al., 2016; Finlay & Read, 1986; McGuire, 2007; Nara, 2006; Newman, 1988; Simard et al., 2012; Teste et al., 2009; but see Brearley et al., 2016).

In order to understand the role of PSFs in natural environments, van der Putten et al. (2016) emphasized the need for field study of potential feedback drivers such as ECM fungi. Ectomycorrhizal fungi form diverse communities in tropical forests and different species may vary in functional traits (Ebenye et al., 2017; Henkel et al., 2012; Johnson et al., 1997). Variation in functional traits could in turn influence the strength of PSFs. For example, some ECM fungal species greatly improve host plant nutrient acquisition, while others have little effect (Nara, 2006). If the ECM fungal community of an adult tree is composed primarily of highly beneficial ECM fungi, then the effect of PSFs on seedlings should be positive. Alternatively, if adult tree ECM fungi impart little benefit to their host tree, positive PSFs should be weak and seedling survival diminished. However, few studies have examined the relationship between positive PSFs and diversity and composition of ECM fungal communities on seedlings (e.g., Nash et al., 2020).

In this study, we examined ECM fungal communities on young seedlings of the TMD tree *Dicymbe corymbosa* Spruce ex Benth. (Fabaceae subfam. Detarioideae) in Guyana. *Dicymbe corymbosa* seedlings established in TMD stands after mast seeding events have high supra-annual survival (Henkel & Mayor, 2019) but do poorly in soils of nearby mixed rainforests (McGuire, 2007). To identify ECM fungal communities that may contribute positively to high seedling survival, we collected seedlings at <1, 6, and 12 months of age. For each seedling, we measured the extent of ECM colonization and isolated ECM morphotypes for high-throughput DNA sequencing. If adult tree ECM fungi improve seedling survival over time, we would expect significant overlap between seedling and adult tree ECM fungal communities. To test this, we compared seedling ECM fungal communities to those of previously sampled nearby adult trees of *D. corymbosa* (Smith et al., 2011, 2017). Because many ECM fungi are host-generalists (Molina et al., 1992), positive effects on seedlings mediated by ECM fungi may occur among heterospecific ECM trees as well. We therefore compared *D. corymbosa* seedling ECM fungi to those of regionally co-occurring heterospecific adult ECM trees (Smith et al., 2011, 2013, 2017). Our primary objective was to characterize ECM fungal communities on TMD seedlings and to observe how these communities change over the course of early seedling development. This is the first study to monitor ECM fungal community assembly over time on seedlings of a TMD tree. It also provides an initial glimpse into possible impacts of ECM fungal community composition on seedling survival in TMD forests.

2 | METHODS

2.1 | Study site

The study was carried out in a *D. corymbosa* forest of the Upper Potaro River Basin in Guyana's Pakaraima Mountains. Annual rainfall averages 3500–4000 mm, with peak rainy seasons in May–July and December–January (Fanshawe, 1952). Daily maximum and minimum temperatures ranged from 25–29 and 19–21°C, respectively (Henkel, unpublished). The site was situated in an undulating valley 20 km east of Mt. Ayanganna and was covered with dense, primarily *Dicymbe*-dominated closed-canopy forest and, to a lesser extent, mixed forests of the *Eschweilera*–*Licania* association (Degagne et al., 2009; Fanshawe, 1952). Study plots were established in *D. corymbosa* monodominant stands within a 5-km radius of a permanent base camp near the Potaro River (5.301389, –59.911111) at elevations of 700–750 m a.s.l. Here, *D. corymbosa* comprises 70–90% of the basal area of canopy trees, heavily dominates the seedling and sapling classes, and no other ECM tree species occur (Henkel, 2003; McGuire et al., 2008). Further details of climate, geology, soils, forest structure, and mast seeding cycles can be found in Henkel (2003), Henkel et al. (2005), Henkel and Mayor (2019), and Henkel et al. (2002). In 2003, five 0.25-ha plots were established (Henkel et al., 2005) and, in 2005, two more 0.25-ha plots were added within a 5-km range of the Potaro base camp (Woolley et al., 2008). All seedlings collected for this study resulted from the 2016 mast seeding event from these seven plots (reported as plots “P1–P7” in Henkel et al., 2005; Henkel & Mayor, 2019).

2.2 | Root sampling

Thirty newly established seedlings per plot were harvested at 6-month intervals between July 2016 and July 2017 following the 2016 mast seeding event. Seedling age classes were defined as: <1-month-old (July 2016; 2–3 weeks after germination), 6-month-old (January 2017), and 12-month-old (July 2017). Whole seedlings were carefully extracted from the soil with their entire root systems (radicle + emerging lateral fine roots) intact, transported to base camp, and their roots rinsed with water. Root system width, depth, and stem height were measured on 6- and 12-month seedlings. Due to time constraints, ECM percent colonization (percent ECM) estimation and morphotype selection for <1-month seedlings were completed in the field (see details below). For 6- and 12-month seedlings, all lateral roots of each seedling were cut from the main root axis and dried with silica gel. Roots were transported to the USA to measure percent ECM, sort morphotypes, and sequence DNA.

2.3 | Seedling mycorrhization

For <1-month seedlings, entire root systems were viewed under a dissecting microscope in the field. We estimated percent ECM of

root systems using the following categories: 0 (0%), 1 (1–25%), 2 (26–50%), 3 (51–75%), and 4 (76–100%). For each 6- and 12-month seedling, the dried lateral roots were cut into 0.5-cm segments and placed in a 10-cm petri dish with a 1 × 1 cm grid. Roots were rehydrated with distilled water and viewed under a dissecting microscope at 40× magnification. Each time a root intersected a gridline it was scored as “ECM” or “non-ECM” based on the presence or absence of a fungal mantle (Giovannetti & Mosse, 1980). Seedling percent ECM was calculated by dividing the number of ECM roots by the total number of root tips scored, multiplied by 100. The categorical scores of <1-month seedlings were converted to percentage scores for comparisons with 6- and 12-month seedlings. For <1-month-old seedlings within each categorical score (0–4), we used the median value of the corresponding percentage range.

Ectomycorrhizal root tips were sorted into morphotypes based on morphological variations. An ectomycorrhiza typically consists of layers of interwoven fungal hyphae (“mantle”) surrounding the root, hyphae penetrating inward from the mantle between the outer root cells to form the interface for nutrient exchange (“Hartig net”), and usually features extramatrical diffuse hyphae and/or hyphal cords extending out from the mantle (Smith & Read, 2008). Observable combinations of color, hyphal density of the mantle, and structure of the extramatrical hyphae may be characteristic of the genus or family of fungus forming the ectomycorrhiza (Agerer, 2006).

2.4 | Illumina sequencing of ECM root tips

Each distinctive morphotype encountered on an individual seedling was placed in cetyltrimethyl ammonium bromide (CTAB). These were grouped by sampling plot and by age class, yielding seven pooled extractions per age class. Each group of pooled ectomycorrhizas was ground in liquid nitrogen, and the DNA extracted using the CTAB method outlined in Vilgalys and Hester (1990). The DNA pellets were resuspended in 50 µl TE (10 mM tris hydrochloride, 1 mM disodium EDTA) and stored at –20°C until use. The ITS1 region was then amplified from each of the extracts using the primer pair ITS1F and ITS2 (Gardes & Bruns, 1993; White et al., 1990). Those primers were amended with a TruSeq 33 bp flanking region (Illumina), so that unique barcodes could be ligated during the next round of PCRs. These PCR products were cleaned with ExoSAP-IT (Thermo Fisher Scientific), and a second PCR was performed using TruSeq DNA amplicon combinatorial dual index adaptors (Illumina). These barcoded libraries were pooled in equal concentrations, along with a barcoded non-biological synthetic control, SynMock (Palmer et al., 2018), and 250-bp paired-end reads were sequenced on one half of a lane on an Illumina MiSeq at the Purdue Genomics Core.

The resulting MiSeq amplicon data were analyzed using the AMPtk pipeline, which processes next-generation sequencing amplicon data from Illumina platforms (Palmer et al., 2018). Forward and reverse primer sequences were trimmed, and paired-end reads were merged using USEARCH v. 9.2.64 (Edgar, 2010). Reads were then clustered into operational taxonomic units (OTUs) at 97% identity

with UPARSE (Edgar, 2013). Possible chimeras were detected and filtered out by the UPARSE-OTU algorithm (Edgar, 2013). To account for index-bleeding, or when reads are assigned to the wrong sample, we used the “filter” command in AMPtk, which filters the OTU table based on the number of reads that bleed into and out of the non-biological synthetic control barcoded sample (Palmer et al., 2018). To increase reproducibility, we only retained OTUs that contained >50 read counts in at least one sample.

Identification of OTUs utilized the global alignment (Lindahl et al., 2013; Selosse et al., 2016) algorithm implemented in USEARCH (Edgar, 2018) on our own custom sequence database, which contains sporocarp and ectomycorrhiza ITS sequences generated over 20 years of ECM sampling in the study area (e.g., Henkel et al., 2012; Smith et al., 2011; Smith et al., 2017; Koch & Aime, *unpublished*). OTUs were considered identified if they shared ≥97% nucleotide identity with an identified sequence in our database. When sequences did not match any samples in the custom database, we used Nucleotide BLAST on GenBank to estimate the closest taxonomic group. The resulting OTU table was curated to remove likely contaminant sequences of taxa that occur exclusively outside of tropical South America. Names were applied to OTUs as follows: (a) binomials for described taxa; (b) genus names with collection numbers for undescribed Guyanese sporocarp morphospecies of T. Henkel (TH) or M.C. Aime (MCA); (c) soil core samples collected in *D. corymbosa* monodominant forest by K. McGuire (e.g., GUYSOILD11, PFG09GadID; McGuire et al., 2010; McGuire et al., 2013); (d) genus name with a clone number (i.e., *Inocybe* 65ms_3g) from a cloning-based dataset (Smith et al., 2017); (e) closest taxonomic group followed by “ecm” and a number (e.g., *Clavulina* ecm1037) or only a number (i.e., *Russula*2) for taxa found only from previous Guyana ECM root tip sampling (Smith et al., 2011, 2017); and (f) closest taxonomic group followed by a unique code (e.g., *Clavulina* i12-194) for taxa found only in this study.

2.5 | Comparisons with adult trees

We determined species overlap of ECM fungi occurring on *D. corymbosa* seedlings and adults of ECM tree species in the Upper Potaro River Basin and the nearby Upper Mazaruni River Basin. Smith et al. (2011) collected 20 ECM roots from each of 19 adult individuals of each of the three sympatric ECM trees *D. corymbosa*, *Dicymbe altsonii* Sandw., and *Aldina insignis* (Benth.) Endl. (Fabaceae subfam. Papilionoideae). This site was located 10 km southeast of the current site in forests with *D. corymbosa* and *D. altsonii* as canopy co-dominants and *A. insignis* trees scattered throughout at low densities. In a subsequent study, Smith et al. (2013) sampled roots from 20 trees each of the two co-dominant ECM trees *Pakaraimaea dipterocarpacea* Maguire & P.S. Ashton (Cistaceae) and *Dicymbe jenmanii* Sandw. in a savanna-fringing forest in the Upper Mazaruni River Basin, located 25 km west of the current study site. Of these five tree species, only *D. corymbosa* forms extensive, purely monodominant stands in the region. In both studies, host tree roots

were traced from the base of the tree to the fine roots, where root samples were collected, and individual ECM root tips were Sanger sequenced. Smith et al. (2017) collected 80 soil cores from different soil microhabitats of ten adult *D. corymbosa* from plots in close proximity to our seedling sampling sites. In that study, 100 healthy ectomycorrhizas were randomly selected from each soil sample and pyrosequenced with 454 technology.

GenBank accession numbers of ECM fungal OTUs from these adult trees were obtained from tables in Smith et al. (2011, 2013, 2017). Our seedling ECM fungal OTUs were BLAST searched to look for conspecific matches with adult tree accession numbers. Adult and seedling root tip OTUs were considered conspecific if they shared at least 97% sequence identity. We then calculated the proportion of *D. corymbosa* seedling OTUs that were conspecific with OTUs recovered from each of the adult ECM tree species. Additionally, datasets for adult *D. corymbosa* were segregated into those occurring near the seedling sampling sites (Smith et al., 2017) and those at 10 km distant (Smith et al., 2011). The proportion of ECM fungi shared with seedlings was calculated for each.

2.6 | Statistical analyses

Statistical analyses were carried out using R software v. 1.1.456 (R Core Team, 2016). Percent ECM and the number of morphotypes per seedling were compared among seedling age classes using non-parametric Kruskal–Wallis rank sum tests, as the data were not normally distributed (Kruskal & Wallis, 1952). Post hoc Mann–Whitney U-tests were performed when rank sum tests showed significant differences between groupings. Boxplot comparisons were generated using the ggplot2 package (Wickham, 2016). Means and standard deviations of percent ECM and the number of morphotypes per seedling among the seven sampling plots were calculated for each age class. OTU accumulation curves were calculated over collection sites for 6 and 12-month seedlings (Figure S1). A correlation matrix was generated to examine relationships between percent ECM, the number of morphotypes per seedling, root system width, root system depth, seedling height, and the number of root tips per seedling using combined data from 6- and 12-month seedlings (Figure S2).

Ectomycorrhizal fungal species richness was estimated for each plot and seedling age class based on the number of unique OTUs. A non-metric multidimensional scaling (NMDS) ordination was created using the metaMDS function in vegan (Oksanen et al., 2019) to visualize groupings of ECM fungal communities based on seedling age class or sampling plot within a seedling age class (Kruskal, 1964). The NMDS ordination used a presence–absence dataset of OTUs assigned to sequences from 6- and 12-month-old seedlings. Species turnover was calculated using the vegan package (Oksanen et al., 2019), with seedling age as the time variable and collection plot as the replicate variable. The NMDS ordination and species turnover used data only from 6- and 12-month-old seedlings, as <1-month-old seedlings were not grouped by plot at time of sampling. We used betadisper() to test whether the dispersions of the

two seedling age groups were homogenous. Although betadisper() revealed heterogeneous dispersions ($p = 0.011$), we used a permutational multivariate analysis of variance (PERMANOVA) to test for a difference in ECM fungal communities between 6- and 12-month seedlings. PERMANOVA has been found to be unaffected by heterogeneous dispersions in balanced datasets such as this current study (Anderson & Walsh, 2013). Tests were based on Morisita-Horn dissimilarity distances and 999 permutations.

3 | RESULTS

3.1 | ECM colonization and community assembly

Percent ECM colonization of *D. corymbosa* seedlings differed among seedling age classes ($\chi^2 = 56.7$; $p < 0.001$) (Figure 1). Percent ECM colonization was significantly higher in 6-month compared to <1-month ($p = 0.015$) and 12-month ($p < 0.001$) seedlings; percent ECM was lowest in 12-month compared to <1-month ($p < 0.001$) seedlings. There was high variation in percent ECM within all seedling age classes: <1-month (45.1 ± 25.0), 6-month (48.3 ± 17.2), and 12-month (35.1 ± 18.1).

Overall, we detected 73 distinct seedling ECM fungal OTUs, including 21 ECM fungal OTUs that were not detected in previous studies. The most OTU-rich ECM fungal lineages were /*russula-lactarius* (35 OTUs), /*clavulina* (19 OTUs), and /*tomentella-thelephora*

(14 OTUs) (Figure 2). OTU richness increased from <1 (6 OTUs) to 6 months (47 OTUs) but did not change from 6 to 12 months (48 OTUs). However, ECM fungal community composition did change from 6 to 12 months, as evidenced by high turnover ($\beta_A = 0.80 \pm 0.05$). The PERMANOVA revealed distinct grouping of ECM fungal OTUs at 6 and 12 months ($p = 0.002$, $F = 12.8$), though heterogeneous dispersions were apparent (Figure 3). Beta-diversity of fungal communities among sampling plots was moderate at 6 months ($\beta_w = 0.47 \pm 0.07$), but higher among plots at 12 months ($\beta_w = 0.70 \pm 0.04$). Only four OTUs were detected in all three seedling age classes: *Clavulina sprucei* group MCA3989 (/clavulina), *Craterellus atratus* Yomyart et al. (/cantharellus), *Inocybe lepidotella* Matheny & Aime (/inocybe), and *Russula puiggarii* group 62rm_3c (/russula-lactarius).

The number of ECM morphotypes on seedlings increased over time ($p < 0.001$). Morphotypes per seedling averaged 1.20 ± 0.69 at <1 month, 3.17 ± 1.27 at 6 months, and 3.55 ± 1.47 at 12 months (Figure 4).

3.2 | Comparisons with adult trees

Dicymbe corymbosa seedlings shared ECM fungi with adult *D. corymbosa*, *D. altsonii*, *D. jenmanii*, *A. insignis*, and *P. dipterocarpacea* (Table 1). Seedlings shared 39 OTUs with adult *D. corymbosa* from the nearby plots of Smith et al. (2017); from the more distant Upper Potaro plots of Smith et al. (2011), seedlings shared 26 OTUs with *D. corymbosa*, 24 with *A. insignis*, and 22 with *D. altsonii*, and 14 and 8 OTUs with *D. jenmanii* and *P. dipterocarpacea*, respectively, from the Upper Mazaruni plots of Smith et al. (2013). Only four ECM fungal OTUs (5.5%) were found on each of the five adult host tree species: *Austroboletus rostrupii* (Syd. & P. Syd.) E. Horak (/boletus), *Russula metachromatica* subsp. *tarumaensis* Singer (/russula-lactarius), *Russula myrmecobroma* S.L. Mill., Aime & T.W. Henkel (/russula-lactarius), and *Tylopilus potamogeton* var. *irengensis* T.W. Henkel (/boletus). None of these OTUs were found on seedlings <1 month old. Forty *D. corymbosa* seedling ECM fungal OTUs (54.8%) had been previously detected on adult conspecifics (Smith et al., 2011, 2017). The remaining 32 seedling ECM fungal OTUs (43.8%) have not been previously recovered from ectomycorrhizas of adult ECM trees in the region. Four of these OTUs are known from sporocarps, including *Clavulina pakaraimensis* Uehling, T.W. Henkel & Aime (/clavulina), *Cortinarius* aff. *amazonicus* Singer & I.J.A. Aguiar (/cortinarius), *Cortinarius* cf. *galeriniformis* Singer & I.J.A. Aguiar (/cortinarius), and *Lactifluus sulcatipes* ined. (/russula-lactarius) (see Henkel et al., 2012). The remaining "unknowns" were 21 OTUs never detected on root tips or from sporocarps, predominantly from the /*tomentella-thelephora* (8), /*russula-lactarius* (5), and /*sebacina* (5) lineages, and seven environmental OTUs in ECM groups detected in soil cores by McGuire et al. (2010) and McGuire et al. (2013). With the exception of *P. dipterocarpacea*, the proportion of seedling ECM fungi in a given age class shared with an adult tree species decreased over 12 months (Figure 5).

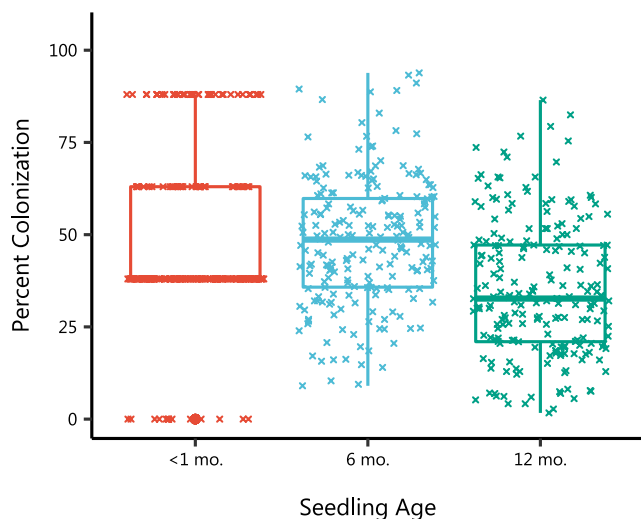


FIGURE 1 Percent ectomycorrhizal fungal colonization of seedling fine lateral roots of the tropical monodominant tree *Dicymbe corymbosa* in Guyana. Seedlings were collected <1, 6, and 12 months following a 2016 mast seeding event. Boxes span the middle 50% of all datapoints, whiskers represent upper and lower quartiles, and median values are represented by thick horizontal bars. Outliers are shown as bold circles and each seedling datapoint is represented by an (x). Percent colonization was significantly higher at 6 months than at <1 month ($p = 0.015$) and 12 months ($p < 0.001$). Percent colonization was also higher at <1 month than at 12 months ($p < 0.001$)

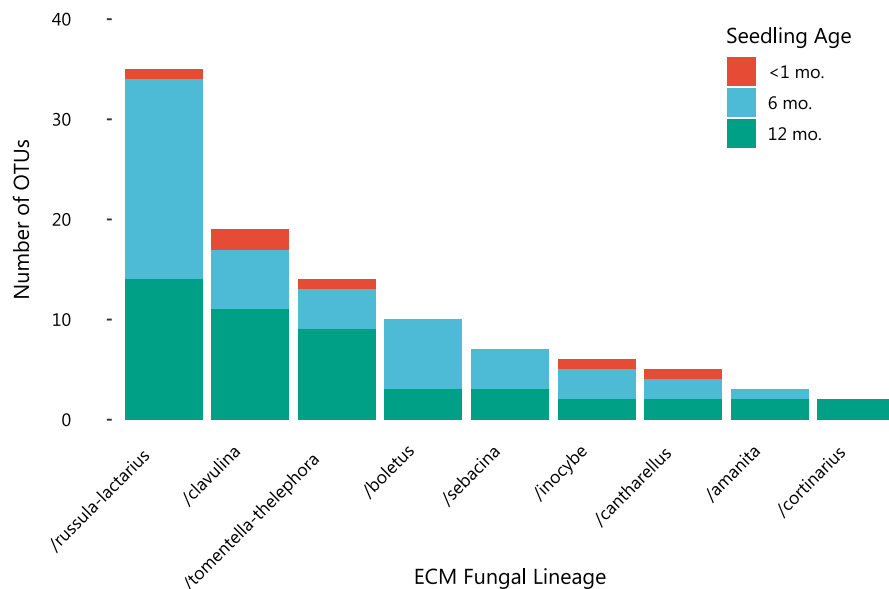


FIGURE 2 Operational taxonomic unit (OTU) richness of ectomycorrhizal fungal lineages defined by Tedersoo et al. (2010) that were recovered from three seedling age groups of the tropical monodominant tree *Dicymbe corymbosa* in Guyana. Ectomycorrhizal fungal OTUs were assigned to sequences generated via Illumina sequencing of pooled seedling root tips. Seedlings were grouped by age at <1, 6, and 12 months following a 2016 mast seeding of *D. corymbosa*

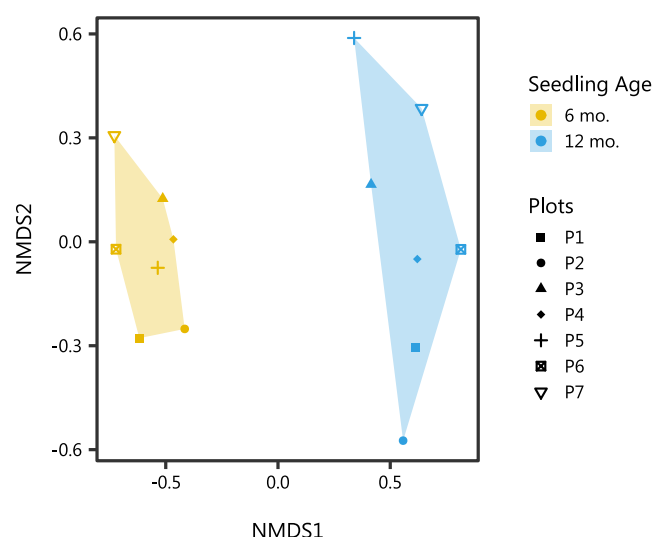


FIGURE 3 Non-metric multidimensional scaling (NMDS) ordination of ECM fungal communities recovered on 6- and 12-month-old seedlings of the tropical monodominant tree *Dicymbe corymbosa* in Guyana. Polygons show grouping of ECM fungal communities based on seedling age, but not on seedling sampling plot. A PERMANOVA test showed that ECM fungal communities between seedling age classes were significantly different ($p < 0.010$). Seedlings <1-month old were not grouped by plot when sampled, so they were excluded from this analysis

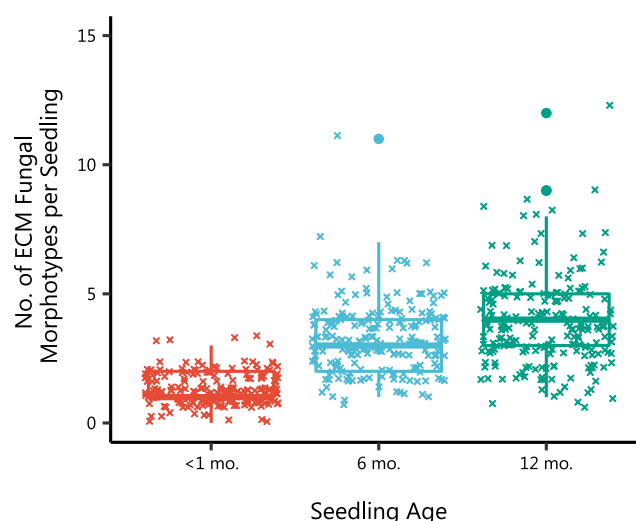


FIGURE 4 The number of visually distinct ectomycorrhizal fungal morphotypes found per seedling of the tropical monodominant tree *Dicymbe corymbosa* in Guyana. Seedlings were collected <1, 6, and 12 months following a 2016 mast seeding event. Boxes span the middle 50% of all datapoints, whiskers represent upper and lower quartiles, and median values are represented by thick horizontal bars. Outliers are shown as bold circles and each seedling datapoint is represented by an (x). There was a significant increase over time in the number of ECM fungal morphotypes per seedling ($p < 0.001$)

4 | DISCUSSION

Over half of the ECM fungal species found over 12 months on *D. corymbosa* seedlings also occurred on nearby adult conspecifics. This is comparable to the Central African monodominant *Gilbertiodendron dewevrei* (Fabaceae subfam. Detarioideae) in which 65% of ECM fungal symbionts were shared between sympatric conspecific seedlings and adults (Ebenye et al., 2017). Similar ECM fungal community composition between seedlings and adults

might indicate adult-seedling mycorrhizal networking, a form of positive PSF that may contribute to early survival of seedlings (Tedersoo et al., 2020). However, Ebenye et al. (2017) utilized carbon and nitrogen isotope data to show that no nutrient transfer occurred between adults and seedlings of *G. dewevrei*. It remains an open question whether networking is occurring in the *Dicymbe* system, as circumstantial evidence for it exists but empirical data are lacking (McGuire, 2007).

TABLE 1 The fifteen most frequently collected ectomycorrhizal (ECM) fungi on *Dicymbe corymbosa* seedling root tips and the presence (+) of these fungi on adults of local ECM tree species in the Upper Potaro (Smith et al., 2011, 2017) and nearby Upper Mazaruni (Smith et al., 2013) River Basins, Guyana. Frequency of seedling ECM operational taxonomic units (OTUs) is the number of plots ($n = 7$ /age class) that an OTU was detected in for both 6- and 12-month-old seedling age classes. Seedlings <1-month-old were not grouped by plot when sampled, so only presence/absence data are shown

Seedling ECM OTUs	Frequency	<1 mo	<i>Dicymbe corymbosa</i> ^a nearby	<i>Dicymbe corymbosa</i> ^b distant	<i>Dicymbe altsonii</i> ^b	<i>Dicymbe jenmanii</i> ^c	<i>Aldina insignis</i> ^b	<i>Pakaraimaea dipterocarpaceae</i> ^c
<i>Russula obtusopunctata</i>	14		+	+	+		+	
<i>Lactifluus subiculatus</i>	12		+					
<i>Russula</i> TH7880	12		+					
<i>Clavulina</i> ecm1037	11		+	+	+		+	+
<i>Russula myrmecobroma</i>	11		+	+	+	+	+	+
<i>Lactarius</i> ecm222	10		+	+	+		+	
<i>Russula</i> aff <i>puiggarii</i> group 62rm_3c	10	+	+					
<i>Clavulina monodiminutiva</i>	9		+	+	+		+	
<i>Clavulina sprucei</i> group MCA3989	9	+	+	+	+	+	+	
<i>Russula metachromatica</i> var <i>tarumaensis</i>	9		+	+	+	+	+	+
<i>Clavulina pakaraimensis</i>	8							
<i>Clavulina sprucei</i> group TH9122	8		+	+				+
<i>Craterellus atratus</i>	8	+	+				+	
<i>Russula</i> cf <i>amnicola</i>	8		+	+			+	
<i>Tomentella</i> ecm974	8			+	+		+	

^aSmith et al. (2017).

^bSmith et al. (2011).

^cSmith et al. (2013).

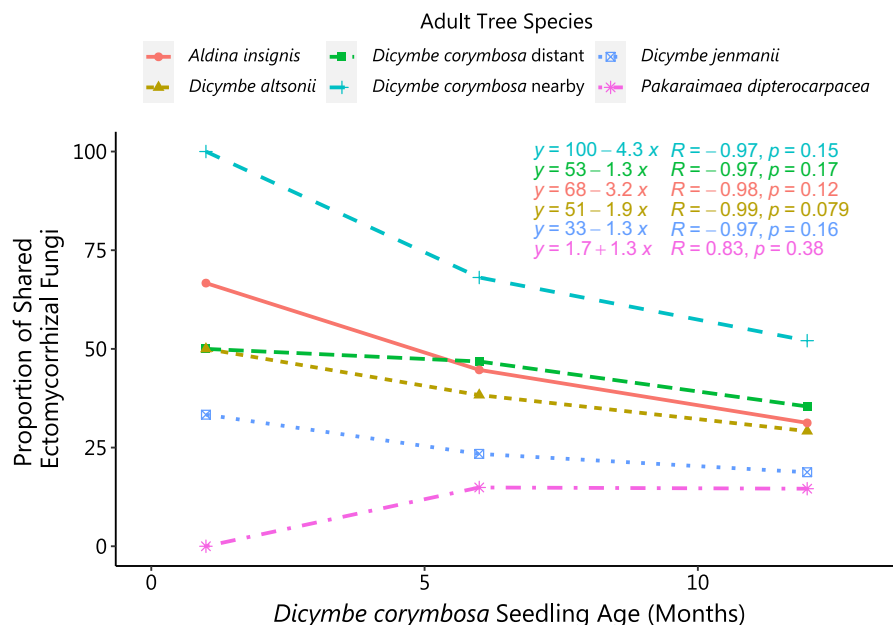


FIGURE 5 Ectomycorrhizal fungal community similarity between seedlings of the tropical monodominant *Dicymbe corymbosa* and adult trees of five regionally co-occurring ECM tree species in Guyana (Smith et al., 2011, 2013, 2017). Seedling ectomycorrhizal root tips were sampled at <1, 6, and 12 months of age and Illumina was used to sequence ITS1 amplicons of all fungi. Community similarity corresponds to the proportion of ectomycorrhizal fungal operational taxonomic units (OTUs) shared with a given adult tree species out of the total number of ectomycorrhizal fungal OTUs detected in that age class. *Dicymbe corymbosa* adults have been separated by distance from seedling sampling sites into nearby *D. corymbosa* (Smith et al., 2017) and more distant *D. corymbosa* (Smith et al., 2011). Linear equations, R, and P values of each linear regression are given in the order they appear in the figure: *D. corymbosa* nearby, *D. corymbosa* distant, *Aldina insignis*, *Dicymbe altsonii*, *Dicymbe jenmanii*, and *Pakaraimaea dipterocarpacea*

Nearly a third of seedling ECM fungi were shared with hetero-specific ECM adult trees, but only one was shared exclusively with a single host species, *Russula* aff. *puiggarii* group ecm720 (*russula-lactarius*) on *A. insignis*. Of seedling ECM fungi shared with adults, most are host generalists, but about a quarter appear to be host-specific, found only on *D. corymbosa* seedlings and adults. In our study system, host specificity may be higher than that documented in mixed dipterocarp-dominant forests of Borneo (Peay et al., 2015), but still less than that seen in hyperdiverse Upper Amazonian forests where ECM tree species occur in very low densities and two-thirds of ECM fungi exhibited host species specificity (Tedersoo et al., 2010). However, *D. corymbosa* adults were sampled with higher intensity using next-generation sequencing, while other regionally sympatric adult tree species were only Sanger-sequenced. Further sampling using the latest technology should be carried out across all ECM host trees in the region to better elucidate host-specific interactions between trees and fungi. Regardless, when only adult tree ECM fungi are considered, no detectable host specificity or distance-decay patterns have been found across host tree species or between sites in Guyana's Upper Potaro Basin (Corrales et al., 2018).

While a majority of seedling ECM fungi were shared with con-specific adults, and to a lesser degree heterospecific adults, nearly half of seedling ECM fungi occurred only on seedlings. These results are similar to those found with the Guinean ECM tree *Cryptosepalum tetraphyllum* where nearly a third of the ECM fungi recovered were found only on seedlings (Diédhiou et al., 2010). In our study, variation in methods used to sample seedlings versus adults may have

contributed to apparently high seedling-specific fungal richness. We exhaustively sampled seedling root tips by collecting entire root systems, an impossible task to perform on adult trees. Contrastingly, with previous adult tree sampling in the region, only the highest quality viable ectomycorrhizas were selected for sequencing (e.g., Smith et al., 2011, 2013, 2017). An additional source of variation may have been the sampling of seedlings and adults in different years, limiting inferences as to whether changes in seedling ECM fungal community structure are preceded by changes in adult ECM fungi. Concurrent sampling could test the hypothesis that adult trees manipulate the soil environment of nearby seedlings and lead to positive PSFs.

The high species turnover of ECM fungi occurring over six months on *D. corymbosa* seedlings may have been influenced by root competition among ECM fungal species (Jonsson et al., 2001; Landeweert et al., 2003; Lilleskov & Bruns, 2003; Wu et al., 1999). Early colonizing ECM fungi on *Pinus* seedlings have been shown to be supplanted over time by different ECM fungal species (Lilleskov & Bruns, 2003; Wu et al., 1999). In contrast, when primary colonizers remain dominant as seedlings age, priority effects may determine ECM fungal species composition (Kennedy & Bruns, 2005). High species turnover of ECM fungal species may also arise from differences in their physiological tolerances, responses to environmental variability over time, and the lifespan of individual fungal genets (Koide et al., 2007). Aside from *P. dipterocarpacea* similarities of seedling-adult fungal communities also decreased with seedling age. Strikingly, the steepest decline in fungal community overlap was between *D. corymbosa*

seedlings and adult conspecifics. Seedling ECM fungal communities were expected to converge on those of adult conspecifics. The observed diversification of ECM fungal symbionts on seedlings might be a unique and important aspect of ECM fungal community assembly in this study system.

We found that percent ECM was not correlated with seedling age. This contrasts with the increase in percent colonization from seedling to sapling size classes seen in *G. dewevrei* (Torti & Coley, 1999). In *D. corymbosa* seedlings, colonization extent and high overall variation in percent ECM (std. dev. = $\pm 21.1\%$) may be influenced by the species composition and interactions of colonizing fungi. Colonization potential may be one of many variable functional traits among ECM fungi. Single-species ECM fungal inoculations of temperate *Pinus* and *Salix* seedlings have experimentally captured the variability of colonization potential, with percent ECM ranging from 5% to 93% depending on the colonizing fungal species (Dosskey et al., 1990; Nara, 2006). Though colonization potential varies among ECM fungi, it does not appear to affect early *D. corymbosa* seedling survival, as seedlings with percent ECM as low as 2% were still viable at 12 months. Abiotic factors can also influence ECM colonization. Percent ECM of *Eucalyptus diversicolor* seedlings was negatively correlated with soil phosphorus (P) concentrations in Australia (Bougher et al., 1990). In Guyana, soil P concentrations drop during masting events (Henkel & Mayor, 2019; Henkel et al., 2005). While it is conceivable that reduced soil P may have triggered a spike in percent ECM in 6-month seedlings, a decrease was seen at 12 months, arguing against this mechanism (Figure 1).

The number of ECM morphotypes per seedling was comparable to other tropical ECM seedlings. Up to 11 ECM morphotypes per 7-month-old seedling were recovered from the dipterocarp *Shorea leprosula* in Malaysia (Alexander et al., 1992), similar to the maximum of 10 for a single 6-month-old *D. corymbosa* seedling (Figure 4). With the Congolian *G. dewevrei*, 19 ECM morphotypes were recovered from a seedling sample of indeterminate age compared to 27 morphotypes from our 6-month *D. corymbosa* seedlings (Torti & Coley, 1999). This same study also found fewer ECM morphotypes per seedling than we did in 6-month seedlings (1.3 ± 0.9 vs. 3.17 ± 1.27). Morphotyping is, however, a subjective method that can under- or overestimate ECM fungal diversity. More accurate comparisons of ECM fungal communities may be facilitated by root tip selection and a combination of sequencing techniques (Menkis et al., 2005). Regardless, we found that individuals and cohorts of *D. corymbosa* seedlings harbored many species of ECM fungi, even though diversity was likely underestimated given that the 97% sequence similarity threshold for assigning conspecificity among ITS1 sequences may not distinguish between members of species complexes (e.g., Henkel et al., 2011).

Ectomycorrhizal colonization of seedlings in this tropical monodominant forest is a dynamic process involving high overall ECM fungal species richness and high species turnover over time. Seedlings were not dependent on extensive colonization, but the diversity of colonizing fungal species may have played an important role in their survival. Seedlings share many ECM fungi with nearby

conspecific and heterospecific adult trees. Additionally, a large portion of mycobionts may be unique to the seedling age class. Positive PSFs have been hypothesized to support growth and survival of seedlings in shaded understories of TMD forests. Such posited mechanisms have placed a heavy emphasis on the role of adult trees in supporting young seedlings through common ECM fungal networks. In the *D. corymbosa* system, adult trees may have networked with very young seedlings, but over the subsequent 12 months seedling ECM fungal assemblages diverged from those of adults. These results suggest that seedlings played a more direct role in assembling their ECM fungal communities than previously thought.

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CONFLICT OF INTEREST

The corresponding author confirms on behalf of all authors that there have been no involvements that might raise the question of bias in the work reported or in the conclusions, implications, or opinions stated.

AUTHOR CONTRIBUTION

CAD: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, resources, software, visualization, writing - original draft. RAK: formal analysis, investigation, methodology, project administration, resources, supervision, writing - review & editing. MCA: funding acquisition, resources, supervision, writing - review & editing. TWH: conceptualization, funding acquisition, investigation, methodology, project administration, resources, supervision, writing - original draft, writing - review & editing.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in the Dryad Digital Repository at <https://doi.org/10.5061/dryad.qz612jmfq> (Delevich et al., 2021) as well as the NCBI Sequence Read Archive under the BioProject PRJNA739004.

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REFERENCES

- Agerer, R. (2006). Fungal relationships and structural identity of their ectomycorrhizae. *Mycological Progress*, 5, 67–107. <https://doi.org/10.1007/s11557-006-0505-x>

- Alexander, I., Ahmad, N., & See, L. S. (1992). The role of mycorrhizas in the regeneration of some Malaysian forest trees. *Philosophical Transactions of the Royal Society B Biological Sciences*, 335, 379–388.
- Alexander, I. J., & Hogberg, P. (1986). Ectomycorrhizas of tropical angiospermous trees. *New Phytologist*, 102, 541–549. <https://doi.org/10.1111/j.1469-8137.1986.tb00830.x>
- Anderson, M. J., & Walsh, D. C. I. (2013). PERMANOVA, ANOSIM, and the Mantel test in the face of heterogeneous dispersions: What null hypothesis are you testing? *Ecological Monographs*, 83, 557–574. <https://doi.org/10.1890/12-2010.1>
- Bagchi, R., Gallery, R. E., Gripenberg, S., Gurr, S. J., Narayan, L., Addis, C. E., Freckleton, R. P., & Lewis, O. T. (2014). Pathogens and insect herbivores drive rainforest plant diversity and composition. *Nature*, 506, 85–88. <https://doi.org/10.1038/nature12911>
- Booth, M. G., & Hoeksema, J. D. (2010). Mycorrhizal networks counteract competitive effects of canopy trees on seedling survival. *Ecology*, 91, 2294–2302. <https://doi.org/10.1890/09-1139.1>
- Bougher, N. L., Grove, T. S., & Malajczuk, N. (1990). Growth and phosphorus acquisition of karri (*Eucalyptus diversicolor* F. Muell.) seedlings inoculated with ectomycorrhizal fungi in relation to phosphorus supply. *New Phytologist*, 114, 77–85.
- Brearley, F. Q., Saner, P., Uchida, A., Burslem, D. F. R. P., Hector, A., Nilus, R., Scholes, J. D., & Egli, S. (2016). Testing the importance of a common ectomycorrhizal network for dipterocarp seedling growth and survival in tropical forests of Borneo. *Plant Ecology & Diversity*, 9, 563–576. <https://doi.org/10.1080/17550874.2017.1283649>
- Comita, L. S., Queenborough, S. A., Murphy, S. J., Eck, J. L., Xu, K., Krishnadas, M., Beckman, N., & Zhu, Y. (2014). Testing predictions of the Janzen-Connell hypothesis: A meta-analysis of experimental evidence for distance- and density-dependent seed and seedling survival. *Journal of Ecology*, 102, 845–856.
- Connell, J. H. (1971). On the role of natural enemies in preventing competitive exclusion in some marine animals and in rain forest trees. In P. J. den Boer & G. R. Gradwell (Eds.), *Dynamics of populations* (pp. 298–310). Center for Agricultural Publications and Documentation.
- Connell, J. H., & Lowman, M. D. (1989). Low-diversity tropical rain forests: Some possible mechanisms for their existence. *The American Naturalist*, 134, 88–119. <https://doi.org/10.1086/284967>
- Corrales, A., Henkel, T. W., & Smith, M. E. (2018). Ectomycorrhizal associations in the tropics – Biogeography, diversity patterns, and ecosystem roles. *New Phytologist*, 220, 1076–1091. <https://doi.org/10.1111/nph.15151>
- Corrales, A., Mangan, S. A., Turner, B. L., & Dalling, J. W. (2016). An ectomycorrhizal nitrogen economy facilitates monodominance in a neotropical forest. *Ecology Letters*, 19, 383–392. <https://doi.org/10.1111/ele.12570>
- Courty, P.-E., Buée, M., Diedhiou, A. G., Frey-Klett, P., Le Tacon, F., Rineau, F., Turpault, M.-P., Uroz, S., & Garbaye, J. (2010). The role of ectomycorrhizal communities in forest ecosystem processes: New perspectives and emerging concepts. *Soil Biology & Biochemistry*, 42, 679–698. <https://doi.org/10.1016/j.soilbio.2009.12.006>
- Degagne, R. S., Henkel, T. W., Steinberg, S. J., & Fox, L. (2009). Identifying *Dicymbe corymbosa* monodominant forests in Guyana using satellite imagery. *Biotropica*, 41, 7–15.
- Delevich, C. A., Koch, R. A., Aime, M. C., & Henkel, T. W. (2021). Data from: Ectomycorrhizal fungal community assembly on seedlings of a Neotropical monodominant tree. Dryad Digital Repository, <https://doi.org/10.5061/dryad.qz612jmfq>
- Diédhiou, A. G., Selosse, M.-A., Galiana, A., Diabaté, M., Dreyfus, B., Bâ, A. M., de Faria, S. M., & Béna, G. (2010). Multi-host ectomycorrhizal fungi are predominant in a Guinean tropical rainforest and shared between canopy trees and seedlings. *Environmental Microbiology*, 12, 2219–2232. <https://doi.org/10.1111/j.1462-2920.2010.02183.x>
- Dosskey, M. G., Linderman, R. G., & Boersma, L. (1990). Carbon-sink stimulation of photosynthesis in Douglas fir seedlings by some ectomycorrhizas. *New Phytologist*, 115, 269–274. <https://doi.org/10.1111/j.1469-8137.1990.tb00452.x>
- Ebenye, H. C. M., Taudière, A., Niang, N., Ndiaye, C., Sauve, M., Awana, N. O., Verbeken, M., De Kesel, A., Séne, S., Diédhiou, A. G., Sarda, V., Sadio, O., Cissoko, M., Ndoye, I., Selosse, M.-A., & Bâ, A. M. (2017). Ectomycorrhizal fungi are shared between seedlings and adults in a monodominant *Gilbertiodendron dewevrei* rain forest in Cameroon. *Biotropica*, 49, 256–267.
- Edgar, R. C. (2010). Search and clustering orders of magnitude faster than BLAST. *Bioinformatics*, 26, 2460–2461. <https://doi.org/10.1093/bioinformatics/btq461>
- Edgar, R. C. (2013). UPARSE: Highly accurate OTU sequences from microbial amplicon reads. *Nature Methods*, 10, 996–998. <https://doi.org/10.1038/nmeth.2604>
- Edgar, R. C. (2018). Accuracy of taxonomy prediction for 16S rRNA and fungal ITS sequences. *PeerJ*, 6, e4652. <https://doi.org/10.7717/peerj.4652>
- Fanshawe, D. B. (1952). *The vegetation of British Guiana: a preliminary review*. Institute paper no. 39. Imperial Forestry Institute.
- Finlay, R. D., & Read, D. J. (1986). The structure and function of the vegetative mycelium of ectomycorrhizal plants. *New Phytologist*, 103, 143–156.
- Gardes, M., & Bruns, T. D. (1993). ITS primers with enhanced specificity for basidiomycetes—Application to the identification of mycorrhizae and rusts. *Molecular Ecology*, 2, 113–118. <https://doi.org/10.1111/j.1365-294X.1993.tb00005.x>
- Giovannetti, M., & Mosse, B. (1980). An evaluation of techniques for measuring vesicular arbuscular mycorrhizal infection in roots. *New Phytologist*, 84, 489–500. <https://doi.org/10.1111/j.1469-8137.1980.tb04556.x>
- Greller, A. M., Gunatilleke, I. A. U. N., Jayasuriya, A. H. M., Gunatilleke, C. V. S., Balasubramaniam, S., & Dassanayake, M. D. (1987). *Stemonoporus* (Dipterocarpaceae)-dominated montane forests in the Adam's Peak Wilderness, Sri Lanka. *Journal of Tropical Ecology*, 3, 243–253.
- Hall, J. S., Harris, D. J., Saltonstall, K., Medjibe, V. P., Ashton, M. S., & Turner, B. L. (2020). Resource acquisition strategies facilitate *Gilbertiodendron dewevrei* monodominance in African lowland forests. *Journal of Ecology*, 108, 433–448.
- Hart, T. B. (1990). Monospecific dominance in tropical rain forests. *Trends in Ecology and Evolution*, 5, 6–11. [https://doi.org/10.1016/0169-5347\(90\)90005-X](https://doi.org/10.1016/0169-5347(90)90005-X)
- Hart, T. B., Hart, J. A., & Murphy, P. G. (1989). Monodominant and species-rich forests of the humid tropics: Causes for their co-occurrence. *The American Naturalist*, 133, 613–633. <https://doi.org/10.1086/284941>
- Henkel, T. W. (2003). Monodominance in the ectomycorrhizal *Dicymbe corymbosa* (Caesalpinaceae) from Guyana. *Journal of Tropical Ecology*, 19, 417–437.
- Henkel, T. W., Aime, M. C., Chin, M. M. L., Miller, S. L., Vilgalys, R., & Smith, M. E. (2012). Ectomycorrhizal fungal sporocarp diversity and discovery of new taxa in *Dicymbe* monodominant forests of the Guiana Shield. *Biodiversity and Conservation*, 21, 2195–2220. <https://doi.org/10.1007/s10531-011-0166-1>
- Henkel, T. W., Aime, M. C., Uehling, J. K., & Smith, M. E. (2011). New species and distribution records of *Clavulina* (Cantharellales, Basidiomycota) from the Guiana Shield. *Mycologia*, 103, 883–894.
- Henkel, T. W., & Mayor, J. R. (2019). Implications of a long-term mast seeding cycle for climatic entrainment, seedling establishment and persistent monodominance in a Neotropical, ectomycorrhizal canopy tree. *Ecological Research*, 34, 472–484. <https://doi.org/10.1111/1440-1703.12014>
- Henkel, T. W., Mayor, J. R., & Woolley, L. P. (2005). Mast fruiting and seedling survival of the ectomycorrhizal, monodominant *Dicymbe corymbosa* (Caesalpinaceae) in Guyana. *New Phytologist*, 167, 543–556.

- Henkel, T. W., Terborgh, J., & Vilgalys, R. J. (2002). Ectomycorrhizal fungi and their leguminous hosts in the Pakaraima Mountains of Guyana. *Mycological Research*, 106, 515–531. <https://doi.org/10.1017/S0953756202005919>
- Hobbie, E. A., & Colpaert, J. V. (2003). Nitrogen availability and colonization by mycorrhizal fungi correlate with nitrogen isotope patterns in plants. *New Phytologist*, 157, 115–126. <https://doi.org/10.1046/j.1469-8137.2003.00657.x>
- Itoh, A., Yamakura, T., Ogino, K., Lee, H. S., & Ashton, P. S. (1997). Spatial distribution patterns of two predominant emergent trees in a tropical rainforest in Sarawak, Malaysia. *Plant Ecology*, 132, 121–136.
- Janzen, D. H. (1970). Herbivores and the number of tree species in tropical forests. *The American Naturalist*, 104, 501–528. <https://doi.org/10.1086/282687>
- Janzen, D. H. (1974). Tropical blackwater rivers, animals, and mast fruiting by the Dipterocarpaceae. *Biotropica*, 6, 69–103. <https://doi.org/10.2307/2989823>
- Johnson, N. C., Graham, J. H., & Smith, F. A. (1997). Functioning of mycorrhizal associations along the mutualism-parasitism continuum. *New Phytologist*, 135, 575–585. <https://doi.org/10.1046/j.1469-8137.1997.00729.x>
- Jonsson, L., Nilsson, M.-C., Wardle, D. A., & Zackrisson, O. (2001). Context dependent effects of ectomycorrhizal species richness on tree seedling productivity. *Oikos*, 93, 353–364. <https://doi.org/10.1034/j.1600-0706.2001.930301.x>
- Kearsley, E., Verbeeck, H., Hufkens, K., Van de Perre, F., Doetterl, S., Baert, G., Beeckman, H., Boeckx, P., & Huygens, D. (2017). Functional community structure of African monodominant *Gilbertiodendron dewevrei* forest influenced by local environmental filtering. *Ecology and Evolution*, 7, 295–304.
- Kennedy, P. G., & Bruns, T. D. (2005). Priority effects determine the outcome of ectomycorrhizal competition between two *Rhizopogon* species colonizing *Pinus muricata* seedlings. *New Phytologist*, 166, 631–638.
- Klironomos, J. N. (2002). Feedback with soil biota contributes to plant rarity and invasiveness in communities. *Nature*, 417, 67–70. <https://doi.org/10.1038/417067a>
- Koide, R. T., Shumway, D. L., Xu, B., & Sharda, J. N. (2007). On temporal partitioning of a community of ectomycorrhizal fungi. *New Phytologist*, 174, 420–429. <https://doi.org/10.1111/j.1469-8137.2007.02000.x>
- Kruskal, J. B. (1964). Nonmetric multidimensional scaling: A numerical method. *Psychometrika*, 29, 115–129. <https://doi.org/10.1007/BF02289694>
- Kruskal, W., & Wallis, W. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47, 583–621. <https://doi.org/10.1080/01621459.1952.10483441>
- Landeweert, R., Veenman, C., Kuyper, T. W., Fritze, H., Wernars, K., & Smit, E. (2003). Quantification of ectomycorrhizal mycelium in soil by real-time PCR compared to conventional quantification techniques. *FEMS Microbiology Ecology*, 45, 283–292. [https://doi.org/10.1016/S0168-6496\(03\)00163-6](https://doi.org/10.1016/S0168-6496(03)00163-6)
- Lilleskov, E. A., & Bruns, T. D. (2003). Root colonization dynamics of two ectomycorrhizal fungi of contrasting life history strategies are mediated by addition of organic nutrient patches. *New Phytologist*, 159, 141–151. <https://doi.org/10.1046/j.1469-8137.2003.00794.x>
- Lindahl, B. D., Nilsson, R. H., Tedersoo, L., Abarenkov, K., Carlsen, T., Kjoller, R., Kõljalg, U., Pennanen, T., Rosendahl, S., Stenlid, J., & Kauserud, H. (2013). Fungal community analysis by high-throughput sequencing of amplified markers – A user's guide. *New Phytologist*, 199, 288–299. <https://doi.org/10.1111/nph.12243>
- Mangan, S. A., Schnitzer, S. A., Herre, E. A., Mack, K. M. L., Valencia, M. C., Sanchez, E. I., & Bever, J. D. (2010). Negative plant-soil feedback predicts tree species relative abundance in a tropical forest. *Nature*, 466, 752–755. <https://doi.org/10.1038/nature09273>
- Mayor, J. R., Schuur, E. A. G., & Henkel, T. W. (2009). Elucidating the nutritional dynamics of fungi using stable isotopes. *Ecology Letters*, 12, 171–183. <https://doi.org/10.1111/j.1461-0248.2008.01265.x>
- McGuire, K. L. (2007). Common ectomycorrhizal networks may maintain monodominance in a tropical rain forest. *Ecology*, 88, 567–574. <https://doi.org/10.1890/05-1173>
- McGuire, K. L., Allison, S. D., Fierer, N., & Treseder, K. K. (2013). Ectomycorrhizal-dominated boreal and tropical forests have distinct fungal communities, but analogous spatial patterns across soil horizons. *PLoS One*, 8, e68278. <https://doi.org/10.1371/journal.pone.0068278>
- McGuire, K. L., Henkel, T. W., Granzow de la Cerda, I., Villa, G., Edmund, F., & Andrew, C. (2008). Dual mycorrhizal colonization of forest-dominating tropical trees and the mycorrhizal status of non-dominant tree and liana species. *Mycorrhiza*, 18, 217–222. <https://doi.org/10.1007/s00572-008-0170-9>
- McGuire, K. L., Zak, D. R., Edwards, I. P., Blackwood, C. B., & Upchurch, R. (2010). Slowed decomposition is biotically mediated in an ectomycorrhizal, tropical rain forest. *Oecologia*, 164, 785–795. <https://doi.org/10.1007/s00442-010-1686-1>
- Menkis, A., Vasiliauskas, R., Taylor, A. F. S., Stenlid, J., & Finlay, R. (2005). Fungal communities in mycorrhizal roots of conifer seedlings in forest nurseries under different cultivation systems, assessed by morphotyping, direct sequencing and mycelial isolation. *Mycorrhiza*, 16, 33–41. <https://doi.org/10.1007/s00572-005-0011-z>
- Miyauchi, S., Kiss, E., Kuo, A., Drula, E., Kohler, A., Sánchez-García, M., & Martin, F. M. (2020). Large-scale genome sequencing of mycorrhizal fungi provides insights into the early evolution of symbiotic traits. *Nature Communications*, 11, 5125.
- Molina, R., Massicotte, H., & Trappe, J. M. (1992). Specificity phenomena in mycorrhizal symbioses: Community-ecological consequences and practical implications. In M. Allen (Ed.), *Mycorrhizal functioning: An integrative plant-fungal process* (pp. 357–423). Chapman and Hall.
- Nara, K. (2006). Ectomycorrhizal networks and seedling establishment during early primary succession. *New Phytologist*, 169, 169–178. <https://doi.org/10.1111/j.1469-8137.2005.01545.x>
- Nash, J., Laushman, R., & Schadt, C. (2020). Ectomycorrhizal fungal diversity interacts with soil nutrients to predict plant growth despite weak plant-soil feedbacks. *Plant and Soil*, 453, 445–458. <https://doi.org/10.1007/s11104-020-04616-y>
- Newman, E. I. (1988). Mycorrhizal links between plants: Their functioning and ecological significance. *Advances in Ecological Research*, 18, 243–270.
- Oksanen, J., Blanchet, F. G., Friendly, M., Kindt, R., Legendre, P., McGlinn, D., Minchin, P. R., O'Hara, R. B., Simpson, G. L., Solymos, P., Stevens, M. H. H., Szoecs, E., & Wagner, H. (2019). *vegan: Community ecology package*. R package version 2.5-6. <https://CRAN.R-project.org/package=vegan>
- Palmer, J. M., Jusino, M. A., Banik, M. T., & Lindner, D. L. (2018). Non-biological synthetic spike-in controls and the AMPtk software pipeline improve microbiome data. *PeerJ*, 6, e4925. <https://doi.org/10.7717/peerj.4925>
- Peay, K. G., Russo, S. E., McGuire, K. L., Lim, Z., Chan, J. P., Tan, S., & Davies, S. J. (2015). Lack of host specificity leads to independent assortment of dipterocarps and ectomycorrhizal fungi across a soil fertility gradient. *Ecology Letters*, 18, 807–816. <https://doi.org/10.1111/ele.12459>
- Phillips, R. E., Brzostek, E., & Midgley, M. G. (2013). Mycorrhizal-associated nutrient economy: A new framework for predicting carbon-nutrient couplings in temperate forests. *New Phytologist*, 199, 41–51.
- Putten, W. H., Bradford, M. A., Pernilla Brinkman, E., Voorde, T. F. J., & Veen, G. F. (2016). Where, when and how plant-soil feedback matters in a changing world. *Functional Ecology*, 30, 1109–1121. <https://doi.org/10.1111/1365-2435.12657>
- R Core Team (2016). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <http://www.R-project.org/>
- Read, J., Hallam, P., & Cherrier, J.-F. (1995). The anomaly of monodominant tropical rainforests: Some preliminary observations in the

- Nothofagus*-dominated rainforests of New Caledonia. *Journal of Tropical Ecology*, 11, 359–389.
- Richards, P. W. (1996). *The tropical rain forest: An ecological study*, 2nd ed. Cambridge University Press.
- Segnitz, R. M., Russo, S. E., Davies, S. J., & Peay, K. G. (2020). Ectomycorrhizal fungi drive positive phylogenetic plant-soil feedbacks in a regionally dominant tropical plant family. *Ecology*, 101, e03083. <https://doi.org/10.1002/ecy.3083>
- Selosse, M.-A., Vincenot, L., & Öpik, M. (2016). Data processing can mask biology: Towards better reporting of fungal barcoding data? *New Phytologist*, 210, 1159–1164. <https://doi.org/10.1111/nph.13851>
- Simard, S. W., Beiler, K. J., Bingham, M. A., Deslippe, J. R., Philip, L. J., & Teste, F. P. (2012). Mycorrhizal networks: Mechanisms, ecology and modeling. *Fungal Biology Reviews*, 26, 39–60.
- Smith, M. E., Henkel, T. W., Catherine Aime, M., Fremier, A. K., & Vilgalys, R. (2011). Ectomycorrhizal fungal diversity and community structure on three co-occurring leguminous canopy tree species in a Neotropical rainforest. *New Phytologist*, 192, 669–712. <https://doi.org/10.1111/j.1469-8137.2011.03844.x>
- Smith, M. E., Henkel, T. W., Uehling, J. K., Fremier, A. K., Clarke, H. D., & Vilgalys, R. (2013). The ectomycorrhizal fungal community in a Neotropical forest dominated by the endemic dipterocarp *Pakaraimaea dipterocarpacea*. *PLoS One*, 8, e55160. <https://doi.org/10.1371/journal.pone.0055160>
- Smith, M. E., Henkel, T. W., Williams, G. C., Aime, M. C., Fremier, A. K., & Vilgalys, R. (2017). Investigating niche partitioning of ectomycorrhizal fungi in specialized rooting zones of the monodominant leguminous tree *Dicymbe corymbosa*. *New Phytologist*, 215, 443–453.
- Smith, S. E., & Read, D. J. (2008). *Mycorrhizal symbiosis*. Academic Press.
- Tedersoo, L., Bahram, M., & Zobel, M. (2020). How mycorrhizal associations drive plant population and community biology. *Science*, 367, eaba1223. <https://doi.org/10.1126/science.aba1223>
- Tedersoo, L., & Brundrett, M. C. (2017). Evolution of ectomycorrhizal symbiosis in plants. In L. Tedersoo (Ed.), *Biogeography of mycorrhizal symbiosis* (pp. 407–467). Springer.
- Tedersoo, L., May, T. W., & Smith, M. E. (2010). Ectomycorrhizal lifestyle in fungi: Global diversity, distribution, and evolution of phylogenetic lineages. *Mycorrhiza*, 20, 217–263. <https://doi.org/10.1007/s00572-009-0274-x>
- Tedersoo, L., Sadam, A., Zambrano, M., Valencia, R., & Bahram, M. (2010). Low diversity and high host preference of ectomycorrhizal fungi in Western Amazonia, a neotropical biodiversity hotspot. *The ISME Journal*, 4, 465–471. <https://doi.org/10.1038/ismej.2009.131>
- ter Steege, H., Henkel, T. W., Helal, N., Marimon, B. S., Marimon-Junior, B. H., Huth, A., Groeneveld, J., Sabatier, D., Coelho, L. D. S., Filho, D. D. A. L., Salomão, R. P., Amaral, I. L., Matos, F. D. D. A., Castilho, C. V., Phillips, O. L., Guevara, J. E., Carim, M. D. J. V., Cárdenas López, D., Magnusson, W. E., ... Melgaço, K. (2019). Rarity of monodominance in hyperdiverse Amazonian forests. *Scientific Reports*, 9, 13822. <https://doi.org/10.1038/s41598-019-50323-9>
- Teste, F. P., Simard, S. W., Durall, D. M., Guy, R. D., Jones, M. D., & Schoonmaker, A. L. (2009). Access to mycorrhizal networks and roots of trees: Importance for seedling survival and resource transfer. *Ecology*, 90, 2808–2822. <https://doi.org/10.1890/08-1884.1>
- Torti, S. D., & Coley, P. D. (1999). Tropical monodominance: A preliminary test of the ectomycorrhizal hypothesis. *Biotropica*, 31, 220–228.
- Torti, S. D., Coley, P. D., & Kursar, T. A. (2001). Causes and consequences of monodominance in tropical lowland forests. *The American Naturalist*, 157, 141–153. <https://doi.org/10.1086/318629>
- van der Heijden, M. G. A., Bardgett, R. D., & van Straalen, N. M. (2008). The unseen majority: Soil microbes as drivers of plant diversity and productivity in terrestrial ecosystems. *Ecology Letters*, 11, 296–310. <https://doi.org/10.1111/j.1461-0248.2007.01139.x>
- van der Velden, N., Ferry Slik, J. W., Hu, Y.-H., Lan, G., Lin, L., Deng, X., & Poorter, L. (2014). Monodominance of *Parashorea chinensis* on fertile soils in a Chinese tropical rain forest. *Journal of Tropical Ecology*, 30, 311–322.
- Vilgalys, R., & Hester, M. (1990). Rapid genetic identification and mapping of enzymatically amplified ribosomal DNA from several *Cryptococcus* species. *Journal of Bacteriology*, 172, 4238–4246. <https://doi.org/10.1128/jb.172.8.4238-4246.1990>
- White, T. J., Bruns, T., Lee, S., & Taylor, J. (1990). Amplification and direct sequencing of fungal ribosomal RNA genes for phylogenetics. In M. A. Innis, D. H. Gelfand, J. J. Sninsky, & T. J. White (Eds.), *PCR protocols: A guide to methods and applications* (pp. 315–322). Academic Press.
- Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*, 2nd ed. Springer.
- Woolley, L. P., Henkel, T. W., & Sillett, S. C. (2008). Reiterations in the monodominant tropical tree *Dicymbe corymbosa* (Caesalpinaceae) and its potential adaptive significance. *Biotropica*, 40, 32–43.
- Wu, B., Nara, K., & Hogetsu, T. (1999). Competition between ectomycorrhizal fungi in colonizing *Pinus densiflora*. *Mycorrhiza*, 9, 151–159.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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